

Supplementary Materials for:

“Dual-Attention Deep Gaussian Processes for Multi-Objective Robust Parameter Design”

A Derivation of the Variational ELBO for Sparse GPs

This appendix provides a detailed derivation of the Evidence Lower Bound (ELBO) (Equation 4) described in Section 2.2. We consider the augmented GP model by introducing M inducing points $\mathbf{Z}$ and corresponding inducing variables $\mathbf{u}$. The joint probability of the observations $\mathbf{y}$, latent functions $\mathbf{f}$, and inducing variables $\mathbf{u}$ is given by:

$$p(\mathbf{y}, \mathbf{f}, \mathbf{u}) = p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\mathbf{u})p(\mathbf{u}), \quad (\text{A.1})$$

where $p(\mathbf{u}) = \mathcal{N}(\mathbf{0}, \mathbf{K}_{\mathbf{uu}})$ is the prior over inducing variables, and $p(\mathbf{f}|\mathbf{u})$ follows the standard GP conditional distribution. To approximate the intractable posterior $p(\mathbf{f}, \mathbf{u}|\mathbf{y})$, we propose a variational distribution $q(\mathbf{f}, \mathbf{u})$. Following the framework of Titsias (2009), we enforce the variational posterior to factorize as:

$$q(\mathbf{f}, \mathbf{u}) = p(\mathbf{f}|\mathbf{u})q(\mathbf{u}), \quad (\text{A.2})$$

where $q(\mathbf{u}) = \mathcal{N}(\mathbf{m}, \mathbf{S})$ is a free variational distribution to be optimized, and $p(\mathbf{f}|\mathbf{u})$ is fixed to the model prior conditionals. This corresponds to Equation (4) in the main text. The goal of variational inference is to minimize the Kullback-Leibler (KL) divergence between the approximate posterior and the true posterior, which is equivalent to maximizing the ELBO:

$$\mathcal{L} = \iint q(\mathbf{f}, \mathbf{u}) \log \frac{p(\mathbf{y}, \mathbf{f}, \mathbf{u})}{q(\mathbf{f}, \mathbf{u})} d\mathbf{f} d\mathbf{u} \quad (\text{A.3})$$

Substituting (A.1) and (A.2) into (A.3), the ELBO can be expanded as:

$$\begin{aligned} \mathcal{L} &= \iint p(\mathbf{f}|\mathbf{u})q(\mathbf{u}) \log \frac{p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\mathbf{u})p(\mathbf{u})}{p(\mathbf{f}|\mathbf{u})q(\mathbf{u})} d\mathbf{f} d\mathbf{u} \\ &= \iint p(\mathbf{f}|\mathbf{u})q(\mathbf{u}) \left[\log p(\mathbf{y}|\mathbf{f}) + \log \frac{p(\mathbf{u})}{q(\mathbf{u})} \right] d\mathbf{f} d\mathbf{u} \end{aligned} \quad (\text{A.4})$$

The term $p(\mathbf{f}|\mathbf{u})$ cancels out in the logarithm ratio, significantly simplifying the computation. The equation splits into two components:

(1) Expected Log-Likelihood:

$$\iint q(\mathbf{u}) p(\mathbf{f} | \mathbf{u}) \log p(\mathbf{y} | \mathbf{f}) d\mathbf{f} d\mathbf{u} = \mathbb{E}_{q(\mathbf{f})}[\log p(\mathbf{y} | \mathbf{f})] = \sum_{i=1}^N \mathbb{E}_{q(f_i)}[\log p(y_i | f_i)], \quad (\text{A.5})$$

where $q(\mathbf{f}) = \int p(\mathbf{f}|\mathbf{u})q(\mathbf{u})d\mathbf{u}$ represents the marginal variational distribution.

(2) KL Divergence:

$$\int q(\mathbf{u}) \left[\int p(\mathbf{f} | \mathbf{u}) d\mathbf{f} \right] \log \frac{p(\mathbf{u})}{q(\mathbf{u})} d\mathbf{u} = -\text{KL}[q(\mathbf{u}) \| p(\mathbf{u})]. \quad (\text{A.6})$$

Note that $\int p(\mathbf{f}|\mathbf{u})d\mathbf{f} = 1$. Combining (A.5) and (A.6), we obtain the final ELBO formulation as presented in Equation (4):

$$\mathcal{L}_{\text{SGP}} = \sum_{i=1}^N \mathbb{E}_{q(f_i)} [\log p(y_i | f_i)] - \text{KL}[q(\mathbf{u}) \| p(\mathbf{u})]. \quad (\text{A.7})$$

B Proof of Unbiased Gradient Estimation via Reparameterization Trick

This appendix provides the mathematical justification for the gradient computation in the Doubly Stochastic Variational Inference (DSVI) framework, specifically concerning Equations (9)-(12) in Section 2.3. We aim to compute the gradient of the expected log-likelihood term in the ELBO with respect to the variational parameters ϕ (i.e., $\mathbf{m}, \mathbf{S}$). The objective function is:

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi(\mathbf{f})} [\log p(\mathbf{y} | \mathbf{f})] = \int q_\phi(\mathbf{f}) \log p(\mathbf{y} | \mathbf{f}) d\mathbf{f}. \quad (\text{B.1})$$

Directly differentiating (B.1) is problematic because the integration domain depends on ϕ . To resolve this, we apply the reparameterization trick defined in Equation (11):

$$\mathbf{f} = g(\phi, \epsilon) = \boldsymbol{\mu}_\phi + \boldsymbol{\Sigma}_\phi^{1/2} \odot \epsilon, \quad \epsilon \sim p(\epsilon) = \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{B.2})$$

Using the Law of the Unconscious Statistician (LOTUS), the expectation over $q_\phi(\mathbf{f})$ can be rewritten as an expectation over the auxiliary noise distribution $p(\epsilon)$, which is independent of ϕ :

$$\mathcal{L}(\phi) = \mathbb{E}_{p(\epsilon)} [\log p(\mathbf{y} | g(\phi, \epsilon))]. \quad (\text{B.3})$$

Now, the gradient operator ∇_ϕ can be moved inside the expectation:

$$\nabla_\phi \mathcal{L}(\phi) = \nabla_\phi \int p(\epsilon) \log p(\mathbf{y} | g(\phi, \epsilon)) d\epsilon = \mathbb{E}_{p(\epsilon)} [\nabla_\phi \log p(\mathbf{y} | g(\phi, \epsilon))]. \quad (\text{B.4})$$

This expectation can be efficiently approximated using Monte Carlo sampling. For S samples of $\epsilon_s \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\nabla_\phi \mathcal{L}(\phi) \approx \frac{1}{S} \sum_{s=1}^S \nabla_\phi \log p(\mathbf{y} | g(\phi, \epsilon_s)). \quad (\text{B.5})$$

Since $g(\phi, \epsilon)$ is a deterministic differentiable function, the gradients can be computed via standard backpropagation (chain rule). This proves that the reparameterization trick provides an unbiased low-variance gradient estimator for optimizing the ELBO in Equation (9).

C Theoretical Justification of the Weighted ELBO

This appendix provides the theoretical basis for the sample-weighted ELBO objective function defined in Equation (16). Standard variational inference minimizes the KL divergence between the variational posterior $q(\mathbf{u})$ and the true Bayesian posterior $p(\mathbf{u}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\mathbf{u})p(\mathbf{u})$. This is equivalent to maximizing the standard ELBO:

$$\mathcal{L} = \mathbb{E}_q[\log p(\mathbf{y} | \mathbf{f})] - \text{KL}[q(\mathbf{u}) \| p(\mathbf{u})]. \quad (\text{C.1})$$

To incorporate the sample-level attention mechanism rigorously, we adopt the framework of generalized variational inference (Knoblauch et al., 2019). In GVI, the standard Bayesian inference is generalized as an optimization problem involving a loss function ℓ , a divergence D , and a prior π :

$$q^* = \arg \min_q \{ \mathbb{E}_q[\ell(\mathbf{y}, \mathbf{f})] + D(q \| \pi) \}. \quad (\text{C.2})$$

In our DA-DGP framework, we define a weighted log-likelihood loss to emphasize the ROI:

$$\ell_{\text{weighted}}(\mathbf{y}, \mathbf{f}) = - \sum_{k=1}^p \sum_{i=1}^N \tilde{s}_{i,k} \log p(y_{i,k} | f_{i,k}^L), \quad (\text{C.3})$$

where $\tilde{s}_{i,k}$ are the normalized attention weights derived in Equation (14). Setting the divergence D to be the KL divergence, the optimization objective becomes:

$$\mathcal{L}_{\text{DA-DGP}} = \sum_{k=1}^p \sum_{i=1}^N \tilde{s}_{i,k} \mathbb{E}_{q(f_{i,k}^L)} [\log p(y_{i,k} | f_{i,k}^L)] - \text{KL}[q(\mathbf{u}) \| p(\mathbf{u})]. \quad (\text{C.4})$$

This derivation confirms that our weighted objective (Equation 16) is not merely a heuristic modification but a valid GVI posterior approximation that minimizes a preference-weighted risk bound.

D Supplementary Visualizations of Experimental Results

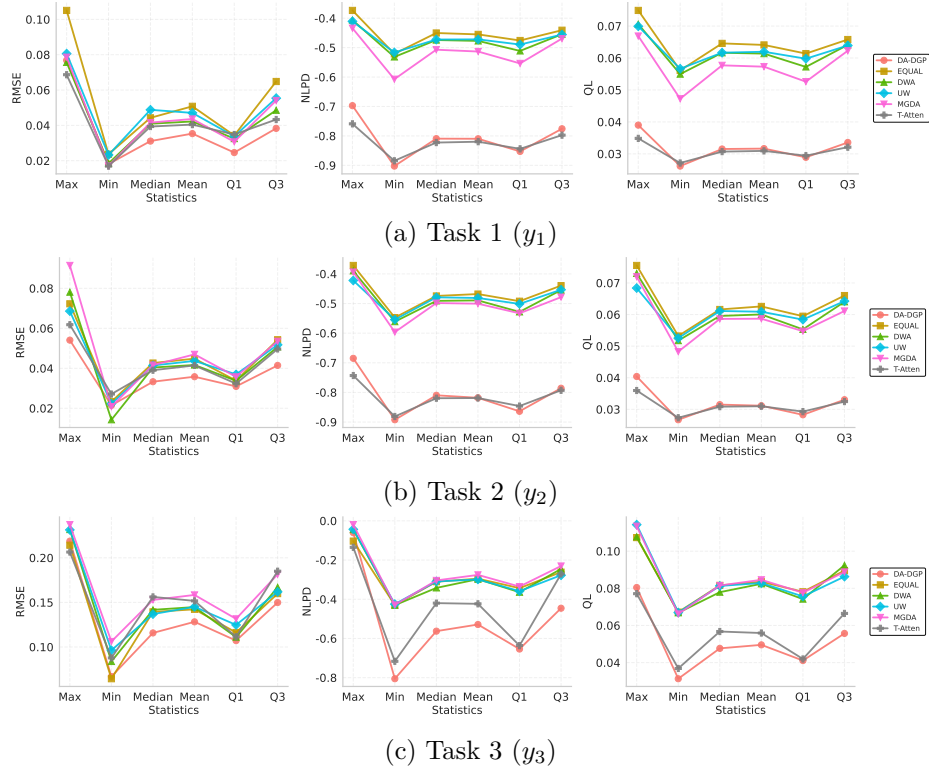


Figure D.1: Performance metric curves for the three tasks.

References

- Titsias, M. (2009). Variational Learning of Inducing Variables in Sparse Gaussian Processes. *Artificial Intelligence and Statistics (AISTATS)*, PMLR, 5, 567–574.
- Knoblauch, J., Jewson, J., & Damoulas, T. (2019). Generalized Variational Inference: Three arguments for deriving new Posteriors. *arXiv preprint arXiv:1904.02063*.